

IHS — Economics

Comprehensive Exam Review Notes

Covering Lectures: Jan 27, Feb 2, Feb 5, Feb 9

Contents

1	Lecture 1 – Introduction to Economics (Jan 27)	3
1.1	What is Economics?	3
1.2	Scarcity — The Fundamental Economic Problem	3
1.3	Specialization and Division of Labor	3
1.4	Production and Distribution	3
1.5	Three Mechanisms for Solving Economic Problems	4
1.6	The Robinson Crusoe Example	4
1.7	Branches of Economics	4
1.8	Key Takeaways – Lecture 1	4
2	Lecture 2 – Economic Agents, Circular Flow & GDP (Feb 2)	5
2.1	Economic Agents	5
2.2	The Circular Flow of Income	5
2.3	Supply Side vs. Demand Side	5
2.4	Gross Domestic Product (GDP)	5
2.5	Three Approaches to Calculating GDP	6
2.6	Real GDP vs. Nominal GDP	6
2.7	GDP Per Capita	6
2.8	Limitations of GDP	6
2.9	Numerical Example	7
2.10	Key Takeaways – Lecture 2	7
3	Lecture 3 – Employment, Fiscal Deficit & Schools of Thought (Feb 5)	8
3.1	Employment and Unemployment	8
3.2	Structural Transformation	8
3.3	Fiscal Deficit	8
3.4	Schools of Economic Thought	8
3.4.1	1. Classical Economics (Adam Smith, David Ricardo)	8
3.4.2	2. Neoclassical Economics	9
3.4.3	3. Marxist Economics (Karl Marx)	9
3.4.4	4. Keynesian Economics (John Maynard Keynes)	9
3.4.5	5. Development Economics	9
3.5	Comparison of Schools of Thought	9
3.6	Key Takeaways – Lecture 3	10
4	Lecture 4 – The Indian Economy (Feb 9)	11
4.1	India’s Economic Journey Since Independence	11
4.2	Economic Growth and Democracy	11
4.3	Poverty Reduction	11
4.4	Employment Challenges: “Jobless Growth”	11
4.5	India’s “Lopsided” Structural Transformation	12
4.6	The Demographic Dividend	12

4.7	Regional Disparities	12
4.8	Labor Force Participation	12
4.9	Key Economic Indicators for India	13
4.10	Key Takeaways – Lecture 4	13
5	Quick-Reference Formula Sheet	14
6	Practice Exam Questions	15
6.1	Multiple Choice	15
6.2	Short Answer Questions	16
6.3	Essay Questions	16
6.4	Problem-Solving	17
7	Glossary of Key Terms	18

1 Lecture 1 – Introduction to Economics (Jan 27)

1.1 What is Economics?

Definition: Economics

Economics is the study of how societies organize themselves to provide for the **material well-being** of their members, focusing on the **production, distribution, and consumption** of goods and services.

Economics addresses a fundamental question: *How do human beings secure their daily bread?* It is concerned not merely with money or finance but with the entire process by which societies produce what they need and decide who gets what.

1.2 Scarcity — The Fundamental Economic Problem

Definition: Scarcity

Scarcity is the condition arising from the **limited availability of resources** (natural, human, and manufactured) to fulfill **unlimited human wants**.

Types of resources:

- **Natural resources** – minerals, water, land, forests
- **Human resources** – labor, skills, knowledge
- **Manufactured resources** – machinery, technology, infrastructure

Because resources are scarce, every society must make *choices* about how to allocate them — this is the heart of the economic problem.

1.3 Specialization and Division of Labor

Definition: Specialization

The process by which individuals or societies focus on producing specific goods or services, leading to greater **efficiency** and **productivity**.

- Specialization allows individuals to develop expertise $\Rightarrow$ higher output.
- It creates **interdependence** — people must trade to obtain goods they do not produce.
- Division of labor separates tasks into distinct occupations and industries.

1.4 Production and Distribution

These are the **two central problems** of any economic society:

1. **Production:** The process of creating goods and services from scarce resources.
2. **Distribution:** How the resulting goods and services are allocated among members of society.

1.5 Three Mechanisms for Solving Economic Problems

Mechanism	How It Works	Examples
Tradition	Roles and tasks determined by custom, social norms, passed through generations	Caste system (India), feudal system (medieval Europe)
Command	Central authority (government/monarch) decides production & distribution	Soviet Union, wartime economies
Market	Individual choices interact via supply & demand; prices serve as signals	Modern capitalist economies

Key Insight

No real-world economy relies on a single mechanism — most are **mixed economies** combining elements of tradition, command, and market.

1.6 The Robinson Crusoe Example

The story of Robinson Crusoe illustrates a **one-person economy**: he must fulfill all needs alone — highlighting the challenges of self-sufficiency and the immense benefits that specialization and trade bring in a larger society.

1.7 Branches of Economics

- Microeconomics
- Macroeconomics
- Development Economics
- Environmental Economics
- Financial Economics
- Economic Growth Theory

1.8 Key Takeaways – Lecture 1

1. Economics studies how societies provide for material well-being.
2. Scarcity is the fundamental economic problem (limited resources, unlimited wants).
3. Specialization and division of labor increase productivity.
4. Production and distribution are the two central economic problems.
5. Tradition, command, and market are the three mechanisms to solve economic problems.
6. Market systems are powerful but can also fail.

2 Lecture 2 – Economic Agents, Circular Flow & GDP (Feb 2)

2.1 Economic Agents

The three main agents in a market economy:

Agent	Role
Households	Consumers of goods & services; suppliers of labor, capital, and entrepreneurship
Firms	Producers of goods & services using inputs (labor, capital, natural resources)
Government	Provides public goods & services; regulates economic activity; redistributes income via taxation & spending

2.2 The Circular Flow of Income

Circular Flow Model

A model showing how income flows continuously between households, firms, and the government through markets for goods & services and markets for factors of production.

How it works:

1. Households supply labor to firms → receive **wages**.
2. Firms produce goods & services → sell to households.
3. Households spend wages on goods & services → revenue flows back to firms.
4. Government collects **taxes** from both households and firms.
5. Government provides **public goods & services** and **transfer payments**.

2.3 Supply Side vs. Demand Side

- **Supply side:** The productive capacity of the economy (firms producing goods & services).
- **Demand side:** The willingness and ability of households, firms, and government to purchase goods & services.

2.4 Gross Domestic Product (GDP)

Definition: GDP

GDP is the monetary value of all **final** goods and services produced within a country's **geographical boundaries** over a given period of time (usually one year).

Important distinctions:

- GDP counts production *within* borders (includes foreigners' production in the country).
- **GNP** (Gross National Product) counts production *by citizens*, regardless of location.
- Only **final goods** are counted to avoid double-counting.

2.5 Three Approaches to Calculating GDP

1. Expenditure Approach

$$\text{GDP} = C + I + G + (X - M)$$

where:

- C = Consumer (household) spending
- I = Investment spending by firms
- G = Government spending
- X = Exports
- M = Imports
- $(X - M)$ = Net exports

2. Production (Value-Added) Approach

$$\text{GDP} = \sum (\text{Value of Output} - \text{Value of Intermediate Goods})$$

Sum the **value added** at each stage of production across all firms to avoid double-counting.

3. Income Approach

$$\text{GDP} = \text{Wages} + \text{Rents} + \text{Interest} + \text{Profits}$$

Sum all incomes earned by factors of production.

Key Insight

All three approaches should theoretically yield the **same GDP figure**, since total expenditure = total production = total income.

2.6 Real GDP vs. Nominal GDP

Nominal GDP	Measured at <i>current</i> prices; does not adjust for inflation.
Real GDP	Measured at <i>constant</i> (base-year) prices; adjusts for inflation — more accurate measure of economic growth.

2.7 GDP Per Capita

GDP Per Capita

$$\text{GDP Per Capita} = \frac{\text{GDP}}{\text{Population}}$$

Provides a rough measure of the **average standard of living** in a country.

2.8 Limitations of GDP

1. Does **not** account for **non-monetary transactions** (e.g., household work, volunteering).
2. Ignores **environmental degradation** and costs.

3. Does **not** measure **income inequality**.
4. Excludes the **informal economy** and black-market activity.
5. Does not capture **quality of life** factors (leisure, health, happiness).

Alternative measures:

- **Human Development Index (HDI)** – combines income, education, and life expectancy.
- **Gini Coefficient** – measures income inequality.

2.9 Numerical Example

Practice Problem

An economy produces 500 units of a good at \$100 each. Households buy 300 units, firms buy 100 units (investment), government buys 100 units.

Solution (Expenditure Approach):

$$\text{GDP} = C + I + G = (300 \times 100) + (100 \times 100) + (100 \times 100) = \$50,000$$

2.10 Key Takeaways – Lecture 2

1. Three economic agents: households, firms, government.
2. Circular flow shows how income moves between agents.
3. GDP = total value of final goods & services within borders.
4. Three approaches: expenditure ($C + I + G + X - M$), production (value added), income (wages + rents + interest + profits).
5. Real GDP adjusts for inflation; nominal does not.
6. GDP has significant limitations — supplement with HDI, Gini, etc.

3 Lecture 3 – Employment, Fiscal Deficit & Schools of Thought (Feb 5)

3.1 Employment and Unemployment

Definition: Unemployment Rate

$$\text{Unemployment Rate} = \frac{\text{Number of people looking for work}}{\text{Working-age population (15–64)}} \times 100\%$$

A low rate (1–4%) is generally indicative of a healthy economy.

Types of Unemployment:

1. **Frictional** – temporary unemployment while transitioning between jobs.
2. **Structural** – mismatch between workers' skills and job requirements.
3. **Cyclical** – caused by economic downturns / recessions.

3.2 Structural Transformation

Definition: Structural Transformation

The process by which an economy shifts from being primarily **agrarian** → **industrial** → **service-oriented**, with corresponding shifts in GDP composition and employment distribution.

Key characteristics:

- Share of agriculture in GDP and employment **declines**.
- Share of industry and services **increases**.
- Typically accompanied by **urbanization** and **rising productivity**.
- Example: South Korea transitioned from agrarian → high-tech industrial economy.

3.3 Fiscal Deficit

Definition: Fiscal Deficit

$$\text{Fiscal Deficit} = \text{Total Government Expenditure} - \text{Total Government Revenue}$$

It represents how much the government needs to **borrow** to finance its spending.

- In India, the fiscal deficit is capped at **3% of GDP**.
- Can be financed through **borrowing** or **printing money**.
- Debate: Should governments run deficits to stimulate the economy, or aim for balanced budgets?

3.4 Schools of Economic Thought

3.4.1 1. Classical Economics (Adam Smith, David Ricardo)

- Emphasizes **individual self-interest** and **free markets**.
- The “**invisible hand**” guides economic activity toward social benefit.
- **Labor theory of value**: value determined by labor required to produce a good.
- **Comparative advantage** (Ricardo): countries should specialize in goods they produce at lower opportunity cost.

- **Malthusian theory:** population grows exponentially while food production grows arithmetically $\Rightarrow$ risk of poverty/famine.

3.4.2 2. Neoclassical Economics

- Builds on classical tradition; focuses on **individual rationality** and **utility maximization**.
- Assumes agents are **self-interested** and make **rational** decisions.
- Emphasizes **market equilibrium** through supply and demand.
- Recognizes **market failures** (externalities, information asymmetry, monopoly power).
- More conservative, market-oriented approach.

3.4.3 3. Marxist Economics (Karl Marx)

- Emphasizes **class struggle** between capitalists (bourgeoisie) and workers (proletariat).
- **Alienation:** workers separated from their labor under capitalism $\Rightarrow$ disconnection and dissatisfaction.
- Focus on the **production sphere** and relationships between economic agents.
- **Exploitation of labor:** surplus value extracted by capitalists.
- Critique of capitalism as inherently unstable and exploitative.

3.4.4 4. Keynesian Economics (John Maynard Keynes)

- Emphasizes **government intervention** to stabilize the economy.
- Particularly important during **recessions and depressions**.
- **Counter-cyclical policy:** government should spend more during downturns and less during booms.
- **Multiplier effect:** government spending has a larger impact than the initial amount spent (money circulates and is re-spent).
- **Phillips Curve:** hypothetical trade-off between inflation and unemployment.

3.4.5 5. Development Economics

- Emerged mid-20th century; focuses on **developing countries**.
- Recognizes that developing countries need **different approaches** than developed ones.
- Emphasizes **protectionism**, **infant industry arguments**, and **structural transformation**.
- Considers the role of **institutions**, **history**, and **culture** in shaping economic outcomes.

3.5 Comparison of Schools of Thought

School	Core Idea	Role of Government
Classical	Free markets, invisible hand	Minimal (laissez-faire)
Neoclassical	Rationality, market equilibrium	Limited; correct market failures
Marxist	Class struggle, exploitation	Radical restructuring of ownership
Keynesian	Demand management	Active fiscal & monetary policy
Development	Context-specific strategies	Significant; protect infant industries

Key Insight

No single school of thought is universally applicable or fully “objective.” Each has strengths and weaknesses, and understanding multiple perspectives is essential for analyzing economic problems.

3.6 Key Takeaways – Lecture 3

1. Unemployment rate = people seeking work / working-age population.
2. Three types: frictional, structural, cyclical.
3. Structural transformation: agriculture → industry → services.
4. Fiscal deficit = government expenditure – revenue; India caps at 3% of GDP.
5. Five major schools: Classical, Neoclassical, Marxist, Keynesian, Development.
6. Keynesian multiplier: government spending amplifies economic impact.
7. Phillips Curve: trade-off between inflation and unemployment.

4 Lecture 4 – The Indian Economy (Feb 9)

4.1 India's Economic Journey Since Independence

India at independence (1947):

- Exceptionally poor with very little resources to invest.
- Primarily agrarian economy.
- Required massive development in infrastructure, industry, and human capital.

Key achievement: India has sustained a **high and rising rate of economic growth** alongside **democracy** — an exception rather than the rule globally. Many high-growth countries achieved growth under authoritarian regimes (e.g., China, South Korea in early stages).

4.2 Economic Growth and Democracy

India's Unique Achievement

India is one of the few countries to demonstrate that **democracy and high economic growth can coexist**. This challenges the conventional wisdom that authoritarian regimes are better suited for rapid development.

4.3 Poverty Reduction

- India has made significant progress: proportion below poverty line has **decreased over time**.
- However, a significant proportion still lives in poverty.
- Economic growth alone is insufficient — **targeted policies** are needed for inclusive development.

4.4 Employment Challenges: “Jobless Growth”

Definition: Employment Elasticity of Growth

$$\text{Employment Elasticity} = \frac{\% \text{ change in employment}}{\% \text{ change in GDP}}$$

India's employment elasticity is **low**, meaning GDP growth has **not** proportionally translated into job creation.

Practice Problem

If India's GDP growth rate increases by 2% annually and the employment elasticity of growth is 0.4, the expected change in employment is:

$$\Delta \text{Employment} = 0.4 \times 2\% = 0.8\%$$

Key concerns:

- Prevalence of **one-person enterprises** and self-employment.
- Large **informal sector**.
- Insufficient growth of formal employment, especially in **manufacturing**.

4.5 India’s “Lopsided” Structural Transformation

Typical path	Agriculture →	Manufacturing → Services
India’s path	Agriculture →	Services (bypassing Manufacturing)

Why this matters:

- Manufacturing can absorb a **large workforce** (unlike services which often need skilled labor).
- Without manufacturing growth, millions of agricultural workers lack better employment options.
- India’s GDP composition shifted, but **employment structure has not shifted proportionally**.

4.6 The Demographic Dividend

Definition: Demographic Dividend

The economic benefit arising when a country’s **working-age population** grows faster than its **dependent population** (children and elderly), resulting in a favorable **dependency ratio**.

Definition: Dependency Ratio

$$\text{Dependency Ratio} = \frac{\text{Dependents (age < 15 or > 64)}}{\text{Working-age population (15–64)}}$$

India’s situation:

- India has a **large youth population** — window of opportunity.
- To leverage this dividend, India must invest in **education, healthcare, and job creation**.
- Risk: India may **grow old before it grows rich** if the dividend is not harnessed in time.

4.7 Regional Disparities

- Different Indian states exhibit vastly different levels of economic development.
- Some states have grown rapidly (e.g., Maharashtra, Karnataka); others lag behind.
- Addressing regional inequality is crucial for balanced national development.

4.8 Labor Force Participation

- India’s overall labor force participation rate is **low**, particularly for **women**.
- **U-shaped curve**: Women’s labor force participation tends to *decrease* as GDP per capita initially rises, then *increases* again at higher income levels.
- Low female LFPR represents a massive underutilization of human resources.

4.9 Key Economic Indicators for India

Indicator	Significance
GDP Growth Rate	Measures the pace of economic expansion
Employment Elasticity	How responsive employment is to GDP growth
Dependency Ratio	Ratio of dependents to working-age population
Poverty Rate	Proportion of population below the poverty line
Labor Force Participation Rate	Share of working-age population in the labor market
Sectoral GDP Composition	Share of agriculture, industry, services in GDP

4.10 Key Takeaways – Lecture 4

1. India achieved growth *with* democracy — a rare combination.
2. Poverty has declined but remains a significant challenge.
3. India suffers from “jobless growth” — low employment elasticity.
4. Structural transformation is “lopsided” (agriculture → services, bypassing manufacturing).
5. The demographic dividend is a time-limited opportunity requiring investment in human capital.
6. Regional disparities and low female labor force participation are major concerns.
7. Risk of “aging before wealth” if demographic window is missed.

5 Quick-Reference Formula Sheet

All Key Formulas

$$\text{GDP (Expenditure)} = C + I + G + (X - M) \quad (1)$$

$$\text{GDP (Production)} = \sum (\text{Output Value} - \text{Intermediate Goods Value}) \quad (2)$$

$$\text{GDP (Income)} = W + R + i + \Pi \quad (\text{Wages} + \text{Rents} + \text{Interest} + \text{Profits}) \quad (3)$$

$$\text{GDP Per Capita} = \frac{\text{GDP}}{\text{Population}} \quad (4)$$

$$\text{Fiscal Deficit} = \text{Gov. Expenditure} - \text{Gov. Revenue} \quad (5)$$

$$\text{Unemployment Rate} = \frac{\text{Unemployed \& Seeking Work}}{\text{Working-Age Population}} \times 100\% \quad (6)$$

$$\text{Employment Elasticity} = \frac{\% \Delta \text{Employment}}{\% \Delta \text{GDP}} \quad (7)$$

$$\text{Dependency Ratio} = \frac{\text{Dependents (age} < 15 \text{ or } > 64)}{\text{Working-Age Pop. (15–64)}} \quad (8)$$

6 Practice Exam Questions

6.1 Multiple Choice

1. What is the primary focus of economics?
 - (a) The study of money and finance
 - (b) The study of how societies provide for material well-being ✓
 - (c) The study of political systems
 - (d) The study of technology
2. What are the three main economic agents in a market economy?
 - (a) Households, Firms, International Organizations
 - (b) Households, Firms, Governments ✓
 - (c) Consumers, Producers, Regulators
 - (d) Buyers, Sellers, Middlemen
3. Which approach calculates GDP by summing $C + I + G + (X - M)$?
 - (a) Production approach
 - (b) Income approach
 - (c) Expenditure approach ✓
 - (d) Output approach
4. What does the Phillips Curve represent?
 - (a) A trade-off between inflation and unemployment ✓
 - (b) Relationship between GDP and fiscal deficit
 - (c) Impact of monetary policy on growth
 - (d) Relationship between exports and imports
5. What is India's primary challenge in structural transformation?
 - (a) Moving from agriculture to industry
 - (b) Moving from industry to services
 - (c) Bypassing the manufacturing sector ✓
 - (d) Reducing the dependency ratio
6. The main limitation of GDP as a well-being measure is:
 - (a) It does not account for income inequality
 - (b) It ignores environmental degradation
 - (c) It excludes unpaid household work
 - (d) All of the above ✓
7. Keynesian economics emphasizes:
 - (a) Government intervention to stabilize the economy ✓

- (b) Laissez-faire policies
 - (c) Class struggle
 - (d) Comparative advantage
8. What term describes economic growth without proportional job creation?
- (a) Structural transformation
 - (b) Demographic dividend
 - (c) Jobless growth ✓
 - (d) Fiscal deficit
9. The U-shaped curve relates to:
- (a) GDP growth and inflation
 - (b) Women's labor force participation and GDP per capita ✓
 - (c) Unemployment and fiscal deficit
 - (d) Population growth and food production
10. Neoclassical economics differs from classical economics by focusing more on:
- (a) Class struggle
 - (b) Government planning
 - (c) Individual rationality and market equilibrium ✓
 - (d) Protectionism

6.2 Short Answer Questions

1. Explain the concept of scarcity and its significance in economics.
2. Describe the circular flow of income and the roles of households, firms, and government.
3. What is the difference between Real GDP and Nominal GDP?
4. Explain the concept of comparative advantage and its relevance to international trade.
5. Define the demographic dividend and explain why India may “grow old before it grows rich.”
6. What is fiscal deficit? What is India's fiscal deficit target?
7. Explain the labor theory of value and its connection to both classical and Marxist economics.
8. What is employment elasticity of growth? Why is India's low elasticity a concern?

6.3 Essay Questions

1. Discuss the three mechanisms (tradition, command, market) societies use to solve economic problems. Provide examples and evaluate the effectiveness of each.
2. Explain the three approaches to calculating GDP. Why should they theoretically yield the same result? Discuss the limitations of GDP as a measure of well-being.

3. Compare and contrast at least three schools of economic thought (Classical, Neoclassical, Marxist, Keynesian, Development Economics). What are the strengths and weaknesses of each?
4. Analyze India's "lopsided" structural transformation. Why has India bypassed manufacturing, and what are the economic consequences? Discuss the demographic dividend in this context.
5. Discuss the relationship between economic growth and democracy, using India as a case study. How has India's experience differed from other high-growth countries?
6. Explain the concept of market failure. How do different schools of thought propose to address it?

6.4 Problem-Solving

1. Calculate GDP given: $C = \$100\text{B}$, $I = \$20\text{B}$, $G = \$30\text{B}$, $X = \$15\text{B}$, $M = \$10\text{B}$.

Solution: $\text{GDP} = 100 + 20 + 30 + (15 - 10) = \155B

2. If GDP grows by 3% and employment elasticity is 0.4, what is the expected employment growth?

Solution: $\Delta\text{Employment} = 0.4 \times 3\% = 1.2\%$

3. 100 people in the labor force, 5 unemployed and seeking work. What is the unemployment rate?

Solution: $\text{Unemployment rate} = \frac{5}{100} \times 100\% = 5\%$

7 Glossary of Key Terms

Term	Definition
Economics	Study of how societies organize production, distribution, and consumption of goods & services
Scarcity	Limited resources vs. unlimited wants
Specialization	Focus on producing specific goods/services for greater efficiency
Division of Labor	Separation of tasks into distinct occupations
GDP	Total monetary value of final goods & services produced within a country's borders
GNP	Total monetary value of goods & services produced by a country's citizens (anywhere)
Real GDP	GDP adjusted for inflation (constant prices)
Nominal GDP	GDP at current prices (not adjusted for inflation)
GDP Per Capita	GDP divided by population
Fiscal Deficit	Government expenditure minus government revenue
Unemployment Rate	Percent of working-age population seeking but not finding work
Structural Transformation	Economy shifting from agriculture to industry to services
Demographic Dividend	Economic benefit from a growing working-age population
Dependency Ratio	Ratio of dependents to working-age population
Employment Elasticity	Responsiveness of employment to GDP growth
Opportunity Cost	Value of the next-best alternative forgone
Invisible Hand	Adam Smith's concept that self-interest guides markets to social benefit
Comparative Advantage	Ability to produce at a lower opportunity cost than others
Multiplier Effect	Government spending amplifies the total economic impact
Phillips Curve	Trade-off between inflation and unemployment
Market Failure	Markets failing to allocate resources efficiently
Alienation	Workers' disconnection from their labor (Marx)
HDI	Human Development Index — combines income, education, life expectancy
Gini Coefficient	Measure of income inequality (0 = perfect equality, 1 = perfect inequality)
Jobless Growth	GDP growth without proportional employment increase
U-shaped Curve	Pattern of women's LFPR declining then rising with development

Good luck on your exam!